# QualityRag Activation Research

## Executable reference system and open validation

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## Abstract

We implement a constraint-directed research reference connecting reviewed ConstraintIR, bounded typed RCT search, signed artifact compatibility, exact source custody, local reconstruction, research-memory operations and isolated deterministic proposal agents. A playable lossless palette scene-state codec provides a concrete video baseline with explicit timeline and exact residuals. It is a predictive reference, not a learned semantic codec or FF-RCT parity implementation. The authored synthetic experiments do not establish a semantic compression advantage or independently validated clinical/legal preservation.

## Existing grammar and governance

The existing finite relation grammar is: Relation<A,B> = EDGE<P:A->B> | PATH(Relation<A,M>, Relation<M,B>) | AND(Relation<A,B>, Relation<A,B>). Typed composition, bounded candidate counts and work limits are reused from the existing Semantic Store. A minimum fitting authored labels remains a candidate, not universal semantic truth. The activation DSL executes PRESERVE, TRANSITION, ACTIVATE, VERIFY and FALLBACK with a deterministic trace and no eval.

The PackSpec source in the finite demonstration explicitly states: "Original source custody evidence is required." The real compiler extracts that authored clause into ConstraintIR. The demonstration maps it to EVIDENCE_PRESENCE with min_independent_sources=1. The existing repository policy examples/apx-packspecs/policies/ffi.json (id=apx-training-ffi, version=1.0) declares verbatim-source-custody, discriminating-probe and independent-corroboration, each with min_independent_sources=1. These genuine evidence-presence clauses are not executable video-fidelity guarantees. The review label is deliberately identified as unauthenticated; missing independent evidence keeps admission INCONCLUSIVE. Distinct policy-source and transformed-data hashes are retained.

## Versioned experimental proposals

The protocol template is `activation-research-v1` (research/protocol.json). Its H1-H6 variables and null hypotheses are explicit; it is not a registered experiment until a real corpus is bound and the holdout is independently opened. The typed search proposal uses `rct.typed-operator-grammar/v1.4`; finite agent proposals use `qualityrag.agent-proposal/v1` and explicit patch lists. The video reference uses `qualityrag.scene-palette/v1` and `qualityrag.scene-video/v1`. These versioned proposals do not assert canonical FF-RCT parity or novelty.

## Methods

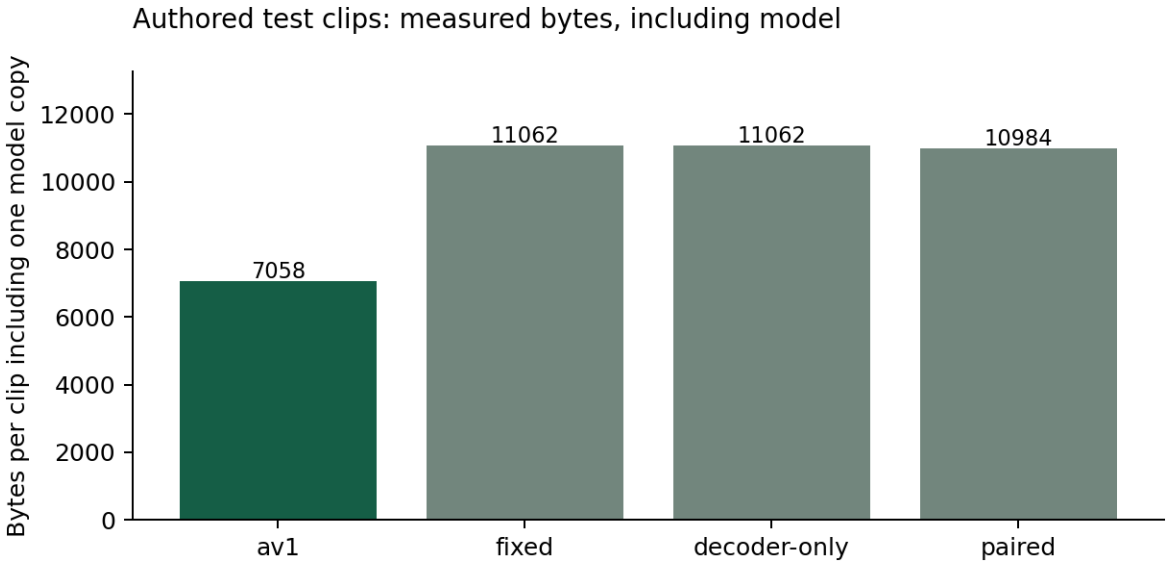
The source-to-output path stores exact original bytes, checks a signed release against runtime, source, policy and transitive modules, reconstructs actual values and checks all declared distinctions. Source adapters retain PDF page or CSV row/line anchors; unsupported image/annotation content remains unknown. Every omission requires a matching policy disposition; unspecified fields are retained rather than silently discarded.

The shared video grammar is a finite RGB backdrop palette. Each frame records its palette code, source frame identity and exact XOR/zlib residual or a raw fallback. Full decoding writes a playable FFV1 file and

verifies normalized RGB frames, presentation timestamps and PCM32 audio samples. The streaming index pins each frame segment and model; range seeking does not fetch the full title.

Six deterministic research roles run in separate resource-limited processes with default-deny seccomp. File opening, sockets and process creation are denied by integration tests. These agents search a predefined typed space; they are not autonomous mathematical inventors. Guardian/counterexample disagreement blocks promotion. Original pinned APX WBC and RCoC are executed through Node 24 rather than reimplemented in Python.

## Observed engineering results




Four authored finite cases reconstruct deterministically. Independent runs produce identical structural receipts. The adversarial suite covers actor/ recipient reversal, negation, temporal order, units, quantities, captions and incompatible models. These finite test outcomes do not authenticate domain truth.

The codec experiment uses two training and two held-out authored clips sharing a synthetic test-pattern generator. This is an engineering fixture, not a representative corpus or four independent natural-video families. The table includes one complete model copy for each clip, a declared cold-install view.

| Arm | Mean measured bytes |
|---|---|
| av1 | 7058 |
| fixed | 11062 |
| decoder-only | 11062 |
| paired | 10984 |

All evaluated reference-arm normalized frames match the source in these clips. The paired palette arm is no smaller than the lossless AV1 baseline. This negative result is retained. Decoder-only retains the fixed encoder/model and selects between scalar and strided-XOR decoders on training data only. No held-out data is sent to the proposing agents. Paired palette selection uses training clips only; the receipt counts its actual search time. No confidence interval or generalization claim is computed from these correlated

authored fixtures. Energy, full lifecycle USD and physical-device effects remain unknown, so no economic advantage is established.

## Falsifiable research protocol

The preregistration template defines H1 bytes, H2 full lifecycle cost, H3 local activation time, H4 preserved distinctions, H5 research continuity and H6 agent evolution. An immutable registry must bind a real corpus and sample plan before an independent evaluator can open its holdout. Family-disjoint splitting, paired bootstrap methods, blinded field scoring and rejection conditions are explicit. The template is not represented as a registered final experiment.

## Claim-to-evidence matrix

| Claim | Reproducible source or receipt | Scope |
|---|---|---|
| Exact finite transformations | tests/test_preservation.py; tests/test_runtime.py | Authored finite scalar records |
| Actual playable lossless reference | tests/test_scene_codec.py | Bounded CFR RGB/PCM32 profile |
| Signed streaming and fallback | tests/test_streaming.py | Pinned local segment store |
| Isolated role proposals | tests/test_agents.py | Linux seccomp deterministic backend |
| Original native APX behavior | tests/test_native_apx.py | Pinned WBC/RCoC source only |
| SDK parity and access isolation | tests/test_service_sdk.py | Same canonical authenticated API |
| Review workflow | tests/test_browser.py | Actual Chromium, opt-in |
| Measured codec byte comparison | engineering-v1.json (SHA-256 5e4ecb3f17ae0783aa3fc4a86cf230e36d2a8c3b2839020afb7c8b4e6805c278) | Authored synthetic engineering run |
| Independent semantic or compression advantage | No evidence supplied | NOT ESTABLISHED |

## Limitations and release decision

A representative authorized 100-clip corpus, licensed neural-model weights, independent domain reviewers, two physical mobile/TV device classes, thermal/ power instrumentation and canonical FF-RCT reconstruction parity are not supplied by this experiment. The neural adapter refuses absent or unpinned weights; it never downloads models or substitutes a pretend result. Lossy neural output cannot qualify for an exact preservation profile.

The release gate is BLOCK with the current evidence. No software test, source hash, agent consensus or ephemeral demonstration signature is treated as independent scientific acceptance. Negative/no-crossover

results remain publishable research outcomes; production promotion is not automatic.

## Reproduction

From the repository root: python -m tools.activation_runtime.run_research --out /tmp/research-run python -m tools.activation_runtime.research.build_report --receipt /tmp/research-run/receipt.json --out /tmp/research-paper

For all tests, install the repository requirements, use Node 24, FFmpeg/ffprobe, and configure QUALITYRAG_APX_ROOT for the pinned native APX source. Browser checks require QUALITYRAG_BROWSER_TESTS=1 plus Playwright Chromium. Test/receipt links must be attached to the actual commit and reviewed before claiming completion.

## Primary references

1. Alliance for Open Media, AV1 specification: https://aomediacodec.github.io/av1-spec/ 2. InterDigital, CompressAI model zoo: https://interdigitalinc.github.io/CompressAI/zoo.html 3. Darwiche and Marquis, A Knowledge Compilation Map, JAIR 17 (2002), https://arxiv.org/abs/1106.1819 4. Doyle, A Truth Maintenance System, MIT AI Memo 521 (1979), https://dspace.mit.edu/entities/publication/5377b306-4ecc-4687-b1f5-78cbb4a0543a 5. Microsoft, GraphRAG indexing/query documentation: https://microsoft.github.io/graphrag/ 6. Generation Is Compression: Zero-Shot Video Coding via Stochastic Rectified Flow (2026), https://arxiv.org/abs/2603.26571 All references were retrieved 2026-11-04. Prior-art and failure boundaries are recorded in research/prior-art.md.